

**IN THE UNITED STATES COPYRIGHT OFFICE
LIBRARY OF CONGRESS**

In re: Section 1201 Study: Notice and Request for Public Comment
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Docket No. 2015-8
80 Fed. Reg. 81369
81 Fed. Reg. 8545 (Extension)

COMMENTS OF CHARLES DUAN

The following comments are respectfully submitted in response to the Notice dated December 29, 2015, for which an extension of the comment period was published February 19, 2016. The commenter is an attorney with the organization Public Knowledge. That organization is submitting comments in this proceeding as well; the present papers are filed in the commenter's personal capacity.

The Notice seeks comments providing "any insights or observations regarding the role and effectiveness of the prohibition on circumvention of technological measures in section 1201(a)." The following comment responds through a short, pertinent work of speculative fiction.

Section 1201, at bottom, asks how enforcement of copyright will influence the development and use of technology. While technological developments offer new opportunities for creativity and free expression, advancements in technology can also lock down that creative process. Striking a balance between the interests of copyright owners and members of the public is the essence of any proper reading of section 1201.

To understand this interplay among technology, creativity, and law, the many factual and analytical comments submitted in this proceeding provide extensive insights into that complex, nuanced problem. But there has always been another way to understand the complexities and nuances of the world: through the device of creative fiction.

Though obviously not based in fact, fiction has been incredibly important in teaching truths. Works like Harriet Beecher Stowe's *Uncle Tom's Cabin*, Upton Sinclair's *The Jungle*, and George Orwell's *1984* have all put forth critical perspectives on important questions of law and policy. Those fictional works made difficult issues of their respective times tangible in a way that no legal document can.

To that end, the following submission is a short story of speculative fiction that deals with the exact question that the Copyright Office considers today with section 1201: how should the protection of copyright interests proceed in the face of new technologies that might be used to stop infringement? The story was published on the popular website *Boing Boing* and was favorably reviewed by well-known writers.

I hope that this story bears interest to those in the Copyright Office who share both a love for creative works and a desire to make good policy in a changing technological environment. I am grateful for the opportunity to submit these comments, and am happy to provide any further information that would be helpful to this proceeding.

Respectfully submitted,

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STOP THE MUSIC

by Charles Duan

Originally published on *Boing Boing*, July 28, 2015

This is a work of fiction. Any resemblance to actual persons or events is purely coincidental.

I.

In the District Court for the Central District of California

Eugene L. Whitman v. Alfred Vail Enterprises, Inc.

Complaint for Copyright Infringement

February 18, 2044

COMES NOW plaintiff, Eugene L. Whitman, by and through his attorneys, and bringing a complaint against Alfred Vail Enterprises, Inc., states:

1. Plaintiff Whitman is a songwriter and composer of the popular song “Taking It Back.”

2. On January 14, 2044, Defendant Vail Enterprises first distributed the now-hit song “Straight Focus.”

3. “Straight Focus” includes an eight-note sample from “Taking It Back.” Vail Enterprises accordingly infringed Whitman’s copyright by making an unauthorized derivative work.

WHEREFORE, Whitman prays for judgment against Vail Enterprises as follows:

A. A permanent injunction prohibiting Vail Enterprises from continuing to infringe Whitman’s copyright;

B. Destruction of all copies of “Straight Focus” in Vail Enterprises’ possession; and

C. An order requiring Vail Enterprises to erase the song “Straight Focus” from the memories of all persons residing in the United States.

II.
In the District Court for the Central District of California
Eugene L. Whitman v. Alfred Vail Enterprises, Inc.
Answer of Alfred Vail Enterprises, Inc.
February 25, 2044

Defendant Alfred Vail Enterprises, Inc., answers the Complaint of Plaintiff Eugene L. Whitman as follows:

1. Vail Enterprises is a corporation of the State of Delaware, with headquarters in Los Angeles, California, and is wholly owned by Alfred Vail.
2. After a long and successful career in the neurobioengineering industry, Mr. Vail decided to enter the field of music. His first composition, “Straight Focus,” was a unique and innovative work of music, reviewed by critics as “the sound of a new technological century” and “a breakthrough piece at the intersection of brain science and art.”
3. Besides being a critical success, “Straight Focus” was a multinational hit. It was viewed on video sharing sites over 350 million times within a month of its release. But more indicative of its popularity are the appreciation videos, remixes, and cover versions that its fans have made.
4. Mr. Vail wrote “Straight Focus” in memory of his daughter, Sarah Vail, who passed away as a teenager due to complications of leukemia last year. The piece is made up of fifty of Ms. Vail’s favorite musical works, which Mr. Vail arranged using his creativity and experience in neurological science to produce an unexpectedly emotional *tour de force* of sound. One of those works was the Plaintiff’s piece “Taking It Back.”
5. The included portion of “Taking It Back” in “Straight Focus” was minimal, and it has not diminished the value of the former piece. Indeed, the popularity of Mr. Vail’s composition has sparked significant interest and increased sales in all of the source pieces. Accordingly, Mr. Vail’s use of a small portion of “Taking It Back” is not copyright infringement, or, at a minimum, is fair use under 17 U.S.C. § 107.

6. Furthermore, Plaintiff Whitman’s proposed remedy of erasing all memory of “Straight Focus” from people’s minds is unprecedented and absurd. No court has ever ordered or authorized a total memory erasure on the public in a copyright case, and it should not be ordered or authorized here.

III.

In the District Court for the Central District of California

Eugene L. Whitman v. Alfred Vail Enterprises, Inc.

Opinion and Order on Memory Erasure

June 20, 2044

OPINION OF BENSON, DISTRICT JUDGE.

The trial jury found that Defendant Vail Enterprises had infringed Plaintiff Whitman’s copyright. Plaintiff now seeks from this Court an order to compel Vail Enterprises to erase all memories of the infringing song “Straight Focus” from the minds of every person residing in the United States, using the National MemSweep system.

This is a highly unusual and unprecedented form of relief that Whitman seeks, and so some background on the system is in order.

The National MemSweep system developed out of the MemSweep technology invented by Alfred Vail in 2028. Prior advances in neuroscience had revealed that human memories could be altered or erased by agitation of brain cells, but those procedures were invasive and risky, accordingly only being used in particularly unusual situations of mental illness such as post-traumatic stress disorder.

Vail discovered that certain rapidly fluctuating low-frequency sound waves could be used to perform the same brain cell agitation, thereby allowing for erasure of memories safely, and noninvasively, and with a great deal of precision as to time and subject matter of the memories. The resulting technology, which he called MemSweep, was offered as a consumer-level service, often to erase memories of embarrassing events, ex-lovers, and traumatic situations.

The consumer service unexpectedly became a national security defense, at the time of the August 2039 terrorist attacks on the United States. The Central Intelligence Agency had intercepted an encrypted communication detailing a plot for multiple simultaneous bombings on several major cities. The CIA knew the attack would happen within a week, but otherwise could not decrypt the messages to identify the details of the plan. With time running out, the CIA in a last-ditch effort procured thousands of large, high-fidelity speaker systems, distributed them around cities, and loudly played MemSweep recordings designed to erase memories of all conversations held during the time of the intercepted messages.

The results were striking: San Francisco and Washington, D.C., where MemSweep was deployed, saw not a single bombing, while New York City, where MemSweep was not deployed in time, was devastated.

Following those attacks, the country moved quickly to deploy MemSweep on a national scale. The resulting array, known as the National MemSweep system, covers every square inch and every person of the United States, and can ensure total erasure of an idea from the populace. The system has been used infrequently but occasionally, and under strict judicial supervision, to foil terrorist plots and avert crimes, to great success, and the CIA and military have investigated applications as well.

But the National MemSweep system has never been used to vindicate a private party. It so far has only been activated to remove ideas of crimes or public dangers. So Whitman's request to use the system to remove memory of a song is truly unexpected. This Court has not a single prior decision or precedent to stand upon.

As an initial matter, an order for a party to use the National MemSweep system would almost never be appropriate, since this Court can only order a party to do something that the party *can* do, and operating National MemSweep is not something that most people can do. But Defendant here is an unusual exception, because Vail Enterprises owns the system. The Alfred Vail who invented MemSweep went on to found Vail Enterprises,

which financed and built National MemSweep, and continues to own it. Thus, an order to use the system could actually be issued against him.

Whitman contends that 17 U.S.C. § 503(b) permits this Court to order Vail Enterprises to perform a memory erasure of the song. That law says that this Court “may order the destruction . . . of all copies or phonorecords found to have been made or used in violation of the copyright owner’s exclusive rights.” Because the neurons storing memories of the song constitute “copies or phonorecords,” Whitman argues, this Court has the power to order Vail Enterprises to conduct the “destruction” of such copies by erasing memories.

I am sympathetic to Whitman’s position. Whitman is clearly very protective of his musical works, refusing to allow anyone else to make derivative works or alterations of them, out of his personal desire to ensure that his music remain “untainted.” He is allowed to do that, as copyright owner. I have already ordered Vail Enterprises to destroy all physical copies of the infringing song.

But it is unclear, in my view, whether ordering the destruction of memories is a bridge too far. Perhaps it is; perhaps it is not—no other court authority provides any guidance on this question. If a Court of Appeals or the Supreme Court were to find such an order permissible, then I would grant the order. But until I have support from a court above, I find it necessary to err on the side of caution and not to order Vail Enterprises to erase memories of the infringing song.

Motion denied.

IV.
The Washington Post
Supreme Court to Hear Case on Memory Erasure
February 12, 2046

This morning, the Supreme Court will hear arguments in a closely watched case over the ability of a songwriter to use his copyright power to erase from the minds of all Americans a song deemed to infringe that copyright.

The case, *Whitman v. Vail Enterprises*, pits songwriter Gene Whitman against neuroscientist-turned-remix-artist Alfred Vail, over the former hit song “Straight Focus.” A jury in 2044 decided that Vail’s song infringed Whitman’s copyright. Immediately afterwards, the Recording Industry Association of America issued commands to the Federal Digital Rights Management system, automatically deleting all versions of the song from all online sites and personal computing devices. “Straight Focus” has not been heard in the United States for over a year now.

But Whitman found the deletion of “Straight Focus” from all devices to be not enough. Intensely protective of his works, Whitman sought a further order from the court for deletion of “Straight Focus” from all people’s minds, using the National MemSweep system owned by Vail.

The trial court refused Whitman’s request, stating that it would not order the use of National MemSweep without guidance from the Supreme Court.

Whitman declined to comment on the case. Vail, in an interview, expressed “exasperation” that the case had even reached the Supreme Court.

“My song ‘Straight Focus’ holds a lot of meaning to lots of people,” he said. “For me, it’s a memory of my daughter, who I lost three years ago. And fans of the song have created their own meanings and memories from it. It boggles the mind why Gene Whitman can selfishly wipe out all those thoughts with some sort of claim of copyright ownership.”

This is the second recent case on the National MemSweep system to reach the Supreme Court. The last case, *United States v. Neilson*, tested the constitutionality of the system as used to suppress criminal activity in the wake of the August 7, 2039 terrorist attacks. A sharply divided 5–4 decision affirmed the system.

Writing for the majority, Chief Justice Diehr rejected challenges under the First, Fifth, and Fourteenth Amendments, finding National MemSweep to be “a necessary tool of a technological society to prevent wrongdoing and malfeasance against the public.” An air force veteran and former Department of Justice prosecutor, the chief justice likely drew

from her years of experience serving in the United States military when she concluded that “the ever-increasing threats to this nation can only be met with ever-increasingly effective defenses.” Furthermore, she wrote, “the duties of citizenship must at least include the surrender of personal thoughts if to do so would protect the greater good, just as one may be called to surrender liberty or property in times of war for the good of the nation.”

A strongly worded dissent by Justice Diamond dismissed the idea that “the human mind is the plaything of the federal government.” Reflecting his background as a former civil rights lawyer, the justice interpreted the Constitution and Bill of Rights to include “penumbras” of guarantees of privacy and freedom of thought. These were strongly in conflict with non-consensual memory erasure, in his view. He cited to the case *Americans for Digital Rights v. Gottschalk*, which held that collection of online data was an illegal search under the Fourth Amendment. Twenty-five years ago, Justice Diamond was the attorney for ADR in that case, which finally struck down pre-Internet caselaw that had been seriously questioned by Justice Sotomayor back in 2012.

Justice Flook, who wrote a separate concurrence, indicated that he was “on the fence” due to the “troubling implications” of widespread memory erasure, but on the whole found that the benefits outweighed those issues. He will likely cast the swing vote in the case, and all eyes will be on him during the argument.

The justice is a former law professor, whose interests and writings focused on environmental and natural resources law. Reflecting his academic prowess and his environmentalist passion, the justice’s opinion in *Neilson*, like many of his others, is both analytically brilliant and emotionally torn. “I sincerely fear,” he wrote, “a world where my memories and the memories of countless others can be erased at the flip of a switch. But I fear attacks of terrorists no less. So long as I can be satisfied that memory erasure is limited to the most necessary of situations, my first fear will remain sufficiently contained.”

The argument will begin at 10:00 AM, and the case is *Whitman v. Alfred Vail Enterprises*, No. 45-405.

V.
In the Supreme Court of the United States
Eugene L. Whitman v. Alfred Vail Enterprises, Inc.
Oral Argument of Attorney Richard A. Tilghman
on Behalf of Petitioner Eugene L. Whitman
February 12, 2046

CHIEF JUSTICE DIEHR: We'll hear argument this morning in Case 45-405, Whitman versus Alfred Vail Enterprises.

Mr. Tilghman?

MR. TILGHMAN: Madam Chief Justice, and may it please the Court.

Today, we ask this Court to protect one of the oldest and most important property rights established within this nation: the copyright that protects the creative works of authors, artists, and—most relevant to this case—musicians.

Section 106 of the Copyright Act guarantees to copyright owners the exclusive right to make copies of their works, to make derivative works based on the originals, and to distribute and publicly perform those works, among other things. These rights are—

JUSTICE DIAMOND: Before we get too far into the copyright weeds, Mr. Tilghman, can you explain to me why we should even be considering this case? This is a case about erasing people's memories, so I want to start out with why you think the National Mem-Sweep system is at all relevant outside the very narrow confines of terrorist activity that we permitted in *Neilson*. I mean, we're talking about suspending civil rights, about freedom of thought. Why should we even be thinking about erasing people's memories anywhere outside the context of terrorism?

MR. TILGHMAN: Justice Diamond, though *Neilson* was itself a case involving a terrorist act, its reasoning was not limited to such situations. This Court's decision relied on general principles of the Constitution, and it decided that erasure of memories on a national scale was permissible when "an unquestionable interest of the American people is placed at imminent risk." The unquestionable interest in *Neilson* was the right to safety from terrorism, but other important interests could fit the bill as well.

CHIEF JUSTICE DIEHR: And, of course, there's the fact that an illegal idea is contraband. You know, no one questions why it's illegal to own illicit drugs, or weapons of mass destruction. An idea can be just as destructive as either of those things. Doesn't it seem like the government should have the power to confiscate dangerous ideas to protect the people?

MR. TILGHMAN: Indeed that's right, your Honor. Possession of an illegal thought ought to be dealt with just like possession of an illegal object. And an infringing copy of a copyrighted work is contraband, as the Chief—

JUSTICE FLOOK: Well—well—hmm. I can understand the contraband argument as a general matter, but I'm worried about the slippery slope. I concurred in *Neilson* but was concerned about what overuse of National MemSweep could ultimately cause. Taken too far, it could lead to censorship, government mind control, *Nineteen Eighty-Four* totalitarianism. How can I be sure, Mr. Tilghman, that what you are asking for in this case won't take us down that path?

MR. TILGHMAN: It's an excellent question, Justice Flook, but luckily it has a straightforward answer. As I said previously, *Neilson* held that an “unquestionable interest of the American people” could justify use of National MemSweep without triggering those slippery slope concerns. And protection of copyright is that kind of unquestionable interest, because copyright is a strong, absolute right.

CHIEF JUSTICE DIEHR: Right, this is the argument from your brief about Digital Rights Management.

MR. TILGHMAN: That's correct.

DRM, of course, is the technology that ensures that copyrighted works cannot be misused in ways that the copyright owners do not want. That technology was once crude and rare, in the early days of computer technology. In that world, perhaps I would have agreed with you, Justice Flook: copyright was rarely enforced, and piracy was rampant.

But the world shifted toward mobile devices around the turn of the century, and those devices could be tracked, controlled, or even disabled remotely. That opened the door to strong, effective DRM, which was ultimately developed by an industry consortium and is now included on every computer device sold today. That industry standard DRM gives copyright owners total control over their works: power to stop copying, power to control who views the work, power even to delete data off of a device to protect against misuse. That is absolute control.

Looking at it that way, what my client is asking for in this case is not a particularly big step. He already has control over every copy of his work stored in any electronic device. All he wants now is control over copies of his work stored in people's minds.

JUSTICE DIAMOND: Wait—wait a second. You're completely forgetting the rights on the other side. Don't consumers have rights to make personal copies, or to play music among friends? There are all sorts of things that are "fair use" or otherwise allowed under copyright law, as I recall. Making parodies, quoting for news reporting, recording broadcasts to watch them later—these are all considered fair use rights, and people are allowed to do them despite copyright.

MR. TILGHMAN: While it is true that those exceptions to copyright still remain on the books, they have all been superseded by the law surrounding DRM.

JUSTICE FLOOK: So to be perfectly honest, Mr. Tilghman, I didn't quite understand this argument in your brief. Section 1201 of the Digital Millennium Copyright Act, which I'm assuming you're talking about, that section says nothing about overriding fair use and other exceptions that Justice Diamond mentioned. So how do you reach the conclusion that Section 1201 supersedes such things?

MR. TILGHMAN: It's admittedly a tricky concept, your Honor, and I apologize if I didn't explain it well in the briefs.

It is true that those exceptions to copyright, like fair use, still remain on the books. But the applicability of those exceptions is sharply limited by DRM and Section 1201. Modern

DRM ensures that protected works cannot be used in ways the author does not desire, even if those uses might fall within one of those exceptions. And Section 1201 made it illegal to get around the DRM. So by operation of law and technology, it is illegal to use a DRM-protected work outside of how the copyright holder permits, regardless of any purported “statutory exceptions” to copyright.

CHIEF JUSTICE DIEHR: So, in other words, Congress made it illegal to do whatever the DRM doesn’t allow, even if the DRM is blocking one of those copyright exceptions. And that means that respecting the DRM is actually more important than those fair use exceptions and such. Is that right?

MR. TILGHMAN: Yes, that’s exactly right. With Section 1201, Congress decided that copyright owners’ interests must come first, when it comes to data on devices. No reason why data in minds should be different.

JUSTICE DIAMOND: That just seems absurd to me. This law makes it illegal for people to exercise their rights like fair use?

MR. TILGHMAN: Perhaps it seems absurd to you that DRM legally overrides fair use, but that has been long established as the law. A case from 2001, *Universal City Studios v. Corley*, specifically said that circumventing DRM was illegal under Section 1201, even if it was done for fair use purposes. *MDY Industries v. Blizzard Entertainment* from 2010 reached the same conclusion.

If these cases were wrong, then Congress has had four decades to change the law. But there has been no such change. Section 1201 remains on the books just as it was enacted. This just goes to show that copyright is a power of total control today. It is a very strong right.

JUSTICE FLOOK: So this is fascinating. I think I now understand your argument for why copyright is a strong right. But let’s step back a minute.

This case is about erasing memories from citizens’ minds. From all the citizens’ minds. Now I appreciate your argument that copyright is a strong right, backed by DRM and laws

like Section 1201. But I'm not sure why I should make the jump from very strong right to this remedy of memory erasure.

Obviously all this MemSweep law is very new, and we're still figuring out how it works in all different areas of the law. But I want to make sure this isn't like the power grabs I remember from the '20s and '30s, when the big nuclear power companies tried to claim all sorts of rights based on some pretty outlandish interpretations of environmental laws. What legitimate need do copyright owners have for erasing people's memories?

MR. TILGHMAN: The legitimate need is the interest in total control over one's copyrighted work. That interest in total control is legitimized by the DRM and Section 1201 protections that enable such control. Copyright law today gives owners total and complete power over their protected property.

The power to control is the power to erase. Already modern DRM systems allow copyright owners to delete infringements of those copyrights from afar at the touch of a button. Physical copies of infringements can be destroyed under Section 503(b) of the Copyright Act. And ever since the U.S. International Trade Commission started treating data transmissions the same as importation of goods in 2014, it has been impounding and blocking Internet information on a regular basis. There is no doubt that removal of information from the public thought is a well-accepted remedy for copyright infringement.

So why should it matter whether that information is removed from a silicon drive or a human neuron? As Chief Justice Diehr explained earlier, information that infringes copyright, like Mr. Vail's song "Straight Focus," is contraband wherever it might be stored. And ultimately, all we ask for in this case is the modest proposal to apply the same power over computer devices that DRM gives copyright owners, to apply that power to human minds. If we can prevent tablets from thinking infringing thoughts, why shouldn't we stop people from using their heads to violate aptly-named intellectual property rights?

The only reason that copyright infringements were allowed to remain stored in people's minds is that we didn't have the technology to remove illegal copies of memories.

Today we have that technology. So I urge the Court to take what is the next natural and reasonable step in protecting copyright owners, to allow them to fully protect what belongs to them.

CHIEF JUSTICE DIEHR: Thank you, Mr. Tilghman.

Ms. Proctor?

VI.

In the Supreme Court of the United States
Eugene L. Whitman v. Alfred Vail Enterprises, Inc.
Oral Argument of Attorney Willa M. Proctor
on Behalf of Respondent Alfred Vail Enterprises, Inc.
February 12, 2046

MS. PROCTOR: Madam Chief Justice, and may it please the Court.

For the almost three centuries that the United States of America has been a nation, the inviolability of the mind has been a central tenet. The pursuit of happiness—the pursuit of personal thought—stands up there with life and liberty as an inalienable right.

But that pursuit of happiness is fundamentally what is being questioned in this Court today. For Petitioner seeks to violate the fundamental right to freedom of thoughts, to maintenance of ideas, to pursuit of happiness—he seeks to violate these in order to vindicate his desire for protection of intellectual property.

Certainly freedom of thought is not absolute, as this Court recognized in *Neilson*. But like freedom of speech, or the rights to due process or equal protection, it is a fundamental right that may be breached only when there is a compelling interest on the other side—an interest like terrorism or national security.

Copyright is no such compelling interest. It does not put the security or protection of the whole nation at stake; it is merely a financial interest of a single person.

CHIEF JUSTICE DIEHR: Well, the interest of a single person can certainly be compelling if that interest is strong enough, can it not? The interest in protecting a person from violent crime, or a person's fundamental right to equal protection under the law,

those are certainly compelling interests. And if, as Mr. Tilghman suggested, copyright is such a strong, absolute right, why shouldn't it fall within that same category of compelling interests deserving of the utmost protection?

MS. PROCTOR: Well that's just it, your Honor. Copyright is not nearly as absolute as Mr. Tilghman would have us believe. The most well-known exception to the absoluteness of copyright is the doctrine of fair use, by which one may perform an act that looks just like infringement of copyright, but bear no liability or responsibility for infringement, because the performed act is deemed a fair and acceptable use.

CHIEF JUSTICE DIEHR: Mr. Tilghman just argued that fair use has basically been trumped by DRM and Section 1201. What's your response to that?

MS. PROCTOR: Maybe he's right as a practical matter, but the annihilation of fair use is not something that this Court should propagate or endorse. Doctrines like fair use are critical to the continued ability of artists and creators to do their work. All art builds upon the successes and inspirations of the past. Just as this Court cites prior cases in writing new opinions, a novel will quote or allude to older works; a painting will use the techniques of the great masters; a musician will borrow ideas from various genres and songs.

So even if DRM and Section 1201 have diminished the doctrine of fair use on electronic devices, this Court should not further diminish it by declaring copyright a totally absolute right. And that is why use of National MemSweep simply cannot be appropriate. That system is reserved for situations of absolutes: absolute rights for which there is no redeeming value on the other side.

JUSTICE DIAMOND: The numerous state laws regulating the overall use of MemSweep technology, those laws would confirm your view that National MemSweep must be reserved for uses of absolute importance, right?

MS. PROCTOR: They certainly would, your Honor. When MemSweep was first made popular years ago, states immediately acted to regulate the industry out of concern that

the technology could be abused. Today, every one of the fifty-two states has laws that license MemSweep operators, restrict use of memory erasure operations to a small set of appropriate situations, impose waiting periods on those wishing to use it, require substantial verification of informed consent—

CHIEF JUSTICE DIEHR: And all of those laws are inapplicable here. As I'm sure you know, the National Security Act of 2040 preempted those state laws and allowed for National MemSweep to be used "to protect any national interest or right," to quote the language of the law. So enforcement of a right created by the U.S. government—oh, say, copyright—is explicitly permitted even in the face of those individual state laws you bring up, right?

Ms. PROCTOR: Your Honor, the National Security Act was enacted only five months after the August 2039 terrorist attacks, and it seems obvious that the act was intended specifically to deal with terrorism and national security. So maybe the text of the act suggests that the act overrides all those protective state laws when it comes to enforcing copyrights. But in the context of the passage of the National Security Act, that's a real stretch of interpretation.

JUSTICE FLOOK: It seems to me, counsel, that what you're pointing to is that larger problem of a slippery slope, which we discussed with Mr. Tilghman. If copyright is not as strong an interest as national security, then allowing the National MemSweep system to be used for copyright enforcement might open the door to all sorts of other unintended uses of the system.

Ms. PROCTOR: That's exactly my concern, your Honor. Abuses of memory erasure—abuses of wholesale national memory erasure—are easy to imagine. A political party in power could use it to weaken the beliefs of the opposing party. Big companies could use it for corporate sabotage. Erasing memories could become the tool of oppression, of ostracism, of—

CHIEF JUSTICE DIEHR: Well it seems to me that there are plenty of other situations where it would be appropriate to use the system. What about leaks of classified information? Shouldn't the government be able to use National MemSweep for preventing leaks?

MS. PROCTOR: Leaks of classified information are still issues of national security, so they are essentially no different from prevention of terrorism. So even if using National MemSweep to prevent those leaks is appropriate, that says nothing about using National MemSweep for copyright infringement.

CHIEF JUSTICE DIEHR: OK, in that case let's consider a copyright example. Say that you have a situation like the old copyright case *Harper & Row v. Nation Enterprises*, where a magazine gets its hands on a book before publication, and "scoops" the book by printing the best parts in advance. That kills the market for the book and is totally contrary to the exclusive right to control distribution of one's work that copyright entails. The only true remedy for the copyright owner is to erase memory of the scoop, so that the book can be sold and read anew. Isn't that an appropriate use as well?

MS. PROCTOR: No, your Honor—

JUSTICE FLOOK: Huh.

MS. PROCTOR: I'm sorry?

JUSTICE FLOOK: Well—hmm. It's interesting, that point that Chief Justice Diehr raises. I knew of the *Harper & Row* case but hadn't thought of it that way.

It reminds me of something that happened to me years ago, when I was still a student in law school. I had been working on a research project on local plastic shopping bag laws, back in 2012. I spent months digging through municipal law records, calling up city councils—even had to visit one town that still only kept the laws in paper books.

Finally, I'd gotten all the data I needed, and was starting to write my paper on it. And I knew that this was going to be a big one, at least for a third-year law student with barely a law review note to his name. But I mentioned the results to a professor of mine at the

time, and he repeated the main conclusion during a press interview. That of course spread across all the news sites within days.

I suppose I should have been happy that the facts were out there. But when it came time to send out my paper for publication two months later, of course no one was interested. The paper was ultimately thrown out, along with the half a year I'd dedicated to it.

Obviously I managed to produce other successful work—

JUSTICE DIAMOND: I think you did pretty well for yourself, Justice Flook.

(Laughter.)

JUSTICE FLOOK: Well, being on the bench with you, it can't be all bad.

(Laughter.)

But I suppose what I take from that incident is—well, having control over your own work is pretty important. I lost control over my research. But now technology can help with that. We have DRM that gives copyright holders control over their work on devices. Maybe that was controversial once, but everyone accepts that today, considering that the Section 1201 law remains unchanged. Why shouldn't we have control over our works in the minds of others? That's all that Mr. Whitman is asking for, right? A sort of do-over, to pluck out the mistaken, infringing information that never should have been let loose in the first place.

CHIEF JUSTICE DIEHR: Something like, maybe, cleaning the environment of a pollutant? That would be analogous.

JUSTICE FLOOK: Hmm—yes—maybe that's it. Maybe that's what I was looking for when I was asking whether memory erasure is the right remedy here. Like scrubbing the air of chemicals, perhaps we're scrubbing minds of information that doesn't belong there.

MS. PROCTOR: I—I see that my time has expired, would—

CHIEF JUSTICE DIEHR: The Court will indulge you a minute or two for a response.

Ms. PROCTOR: Thank you, your Honor. To answer your question, Justice Flook, we're not just scrubbing minds of a small piece of information. We're scrubbing a whole lot more.

When I talked to my client, Mr. Vail, about the importance of his song, he explained that it was intimately tied to the memory of his daughter, Sarah Vail. "Straight Focus" is made up of her favorite songs, he said, and so his memory of that song is his memory of her. The song also uses the neurological techniques he invented, working as a carefully orchestrated whole to trigger memories of Sarah in his mind. In a very real sense, it is this song that keeps Mr. Vail's daughter alive. To take away his memory of that song would be to take a piece of her away from him.

This deplorable result is only writ larger for all the fans of the song. Fans have enjoyed the piece and built their own memories around it. Artists have built upon and improved the song with their personal tastes and creativity through remixes and cover versions.

Shall we give up all of this creation, all of this progress, all of this thought and happiness, on the unilateral request of one songwriter? The Constitution of the United States says that copyright law must "promote the progress of science and the useful arts." But the memory erasure that Mr. Whitman seeks would eliminate not only the infringing song—which was a work of progress in itself—but also all the progress made based upon it. Erasure is regress, not progress, and this Court should not authorize it.

CHIEF JUSTICE DIEHR: Thank you, Ms. Proctor. The case is submitted.

VII.
The Washington Post
High Court Authorizes Erasure of Song from National Memory
June 25, 2046

In an acrimoniously split 5–4 decision, the Supreme Court approved an order against technology and music giant Vail Enterprises, requiring it to use the National MemSweep

system to erase all memory of the hit song “Straight Focus” from the minds of all people in the United States.

Writing for the majority, Justice Flook reiterated his concerns about cabining uses of National MemSweep to “serious infractions of the law,” but was persuaded that “the strong copyright policies that have developed over the last few decades indicate that the nation now views copyright infringement as such a serious infraction.” He went on to find that “removal of memories is an appropriate remedy, given the longstanding ability of copyright owners to remove ideas from just about every other place, based on Section 1201 and other parts of the copyright law.”

In dissent, Justice Diamond found the decision “diametrically opposed to the constitutional requirement that copyrights promote progress of the arts and sciences,” and predicted that the decision would lead to “an era devoid of new music, new writings, and new creations that depend on building upon the work of the past.”

Music industry executives applauded the high court for approving of memory deletion. “Copyright should give creators total control over their works and how audiences perceive them,” said Clifford King, president of the Recording Industry Association of America. “Eliminating bootleg content from people’s minds ought to be a part of that.”

But civil liberties advocates disagreed. “A person’s right to freedom of thought ought to trump a commercial copyright interest,” read a letter signed by twenty nonprofit organizations this morning, asking Congress to overturn the decision.

The ruling will most directly affect the creator of “Straight Focus” itself, Alfred Vail, who will be compelled to conduct the erasure of memories of his own song. Vail could not be reached for comment. His neighbors reported that they had not seen him leave his home since the ruling was announced.

This particular act of memory erasure for copyright infringement will remain unique for some time, given that it could only occur in the unusual circumstance of the copyright

infringer being the same as the owner of the National MemSweep system. But that may not be the case for long. The patents on MemSweep are due to expire two years from now.

“We are in the early stages of planning our own system, tentatively called the ‘Mental Rights Management’ or MRM system,” said Mr. King. “This will allow every author, artist, and songwriter to reap the benefits of memory erasure that the Supreme Court’s decision authorizes.”

The activation of National MemSweep in view of this case will not occur for several weeks, most likely not until late July. System engineers will code in the parameters of the information to be erased, so that no trace of memory—not even knowledge that a memory had been erased—will remain.

As with other activations of the system, it will likely occur in the late morning or early afternoon, to minimize disturbance. The procedure will consist of about fifteen seconds of low-pitched, rhythmic sounds—some have described them as lyrical or soothing—loud enough to penetrate walls or buildings, and calibrated to remove all memories of the song in question. After that time, the music will go silent.

VIII.

Message from Rand A. Warsaw to Alice Stevens Vail

April 4, 2084

Hi Mom,

(I’m trying out this new direct-mind transcription thing—sorry if this is a little rambling, haven’t quite gotten the mental focus down yet.)

I was going through Grandpa Al’s things to get ready for the estate sale next week (I can’t believe how fast they do these things, the funeral was just a week ago!) and of all things, there was this box stuck behind a desk stored in the attic. At first it looked like it might have fallen back there by accident—it must have been there for years, it was so dusty—but I think Grandpa may have meant to hide it.

Inside there were a couple of legal documents and newspaper clippings. Something about a lawsuit over a song that he wrote. I had no idea that he was a musician once. But I guess that's not surprising, considering that the documents say that they erased everyone's memory of his song.

On top of the papers, there was this note:

June 26, 2046. Last Monday, the Supreme Court approved obliterating my song "Straight Focus" from everyone's memory. My song! That I spent months thinking about and planning, all because of a couple of dumb notes borrowed from someone else.

I thought about leaving the country so at least I could escape the MemSweep order, but the judge said that I had to be present to carry out the order to activate the system next month. Besides, the RIAA guy said that it was only a matter of time before their Memory Rights Management system went global anyway, so it's not like I could really get away.

I really hoped that "Straight Focus" would be my lasting legacy. I remember explaining MemSweep to Sarah when she was seven or eight, and she said to me, "Dad, maybe you should make something for remembering, instead of something for forgetting." And after she lost her battle with cancer, I realized I did want to make something for people to remember—and something for me to remember her by.

"Straight Focus" did that. Everyone loved the song, and there were all those great fan versions of it, and videos and everything. I was sharing my new music ideas with the whole world—creating something to be remembered.

Copyright is supposed to protect artists like me, isn't it? And instead, copyright law is what's destroying my creation. How could they not see something like this coming? How could they have left that ridiculous 1201 law untouched for almost fifty years?

Well, life is full of ironies. The last thing I expected was for my own memory erasing technology to be used against my own ideas.

It's been losses all the way down ever since Sarah died. She loved music, and there was nothing more important to her than her favorite playlist. After I lost her, I barely managed to hold onto that playlist before they deleted it under the digital goods estate planning law (bizarre that your next of kin can't inherit your music library!). I made "Straight Focus" to keep her memory alive, but then they deleted that song, even on my own computers.

And now they're deleting the very memory of that song about her. It's like they're deleting her from my mind.

So before my memories are taken from me next month, I'm packing up what I have left of this song—practically all my memories of Sarah, I guess. I was going to leave the box on my desk so I'd see it after they run MemSweep.

But the thoughts I'm having now are too sad, too painful, and I don't know if I want to reopen those wounds right after I've lost my memories. So I'm going to tuck this box back here in the attic. Perhaps I will find it again one day, hopefully in happier circumstances.

Also, there's this plastic cartridge, about the size of my hand, with some sort of shiny brownish spool of tape running inside. The plastic cartridge has "Straight Focus" written on it. There's also a black machine with some buttons, and it looks like the cartridge fits inside the machine. I can't get them to turn on, though. I think they need some power, and I tried every wireless induction charger I've got at home, and none of them worked. Must be too old for them, I guess.

I'll bring all the stuff I found when I come by tomorrow. Let's ask Uncle James if he can figure it out, since he likes all that old 21st-century machinery. Who knows, maybe we'll discover a whole new side of Alfred Vail that we'd never heard before.

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